A Finite-Rate Selection Law for Annealed Gibbs Sampling

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Paper · code · interactive edition · film
github.com/ali-izhar/finite-rate-selection-law

PLAUSIBLE SAMPLES, WRONG PROPORTIONS

In Curie–Weiss and Hopfield models, annealed Gibbs samples keep only a fraction c of the model's log-odds between modes.

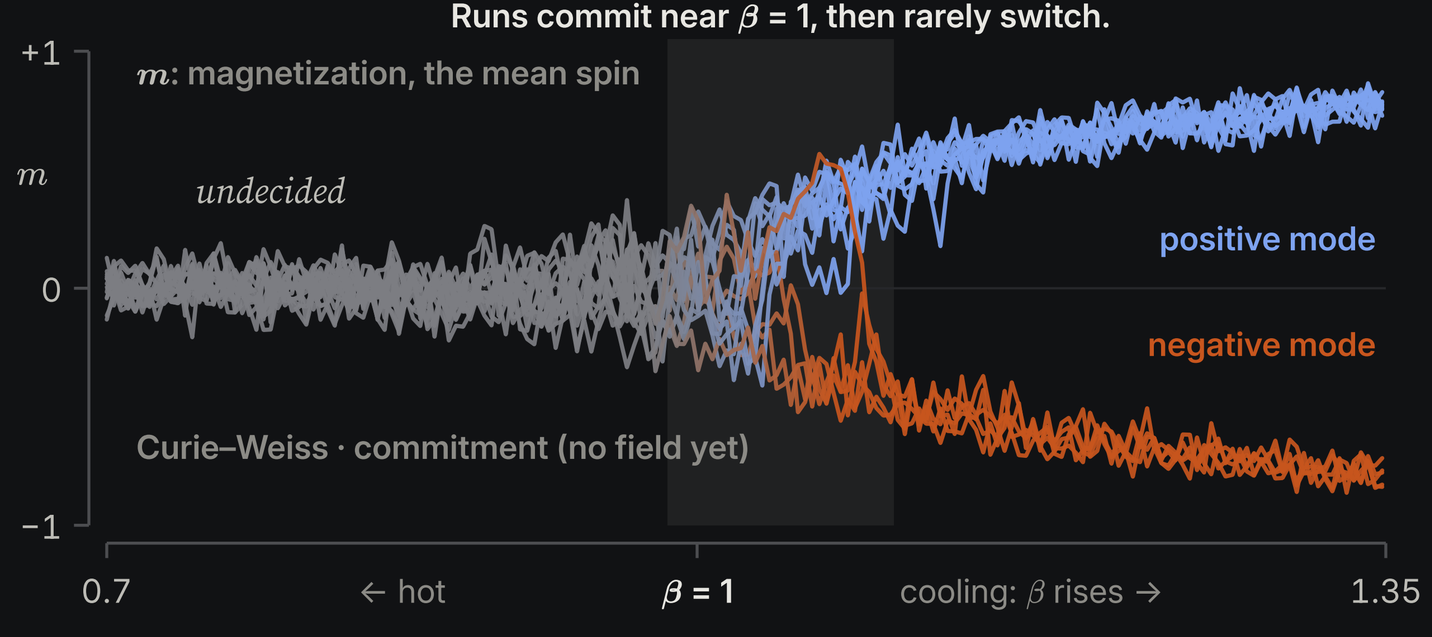


Fig. 1a · 14 paths of the drift-plus-noise approximation · N = 1,024 · G = 0

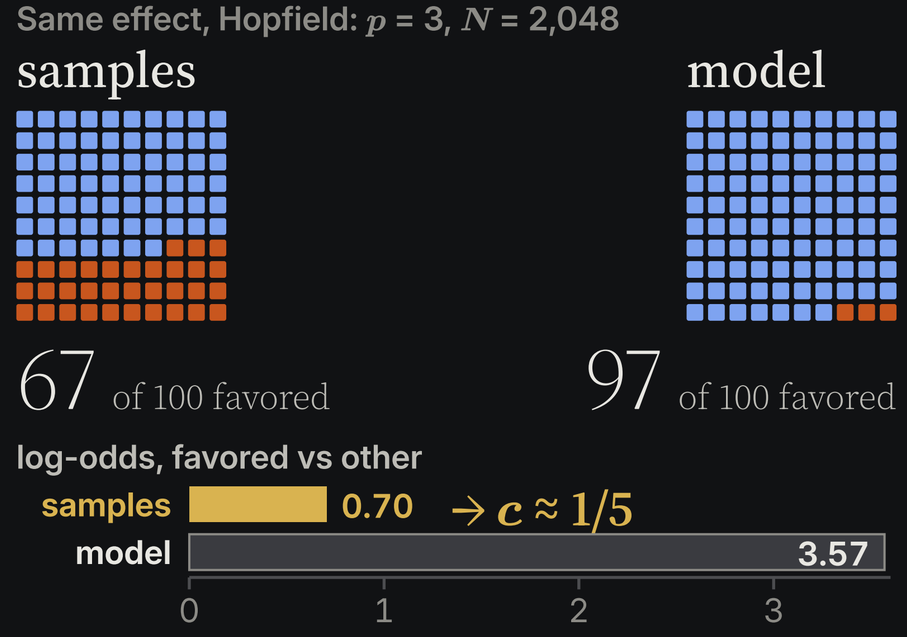
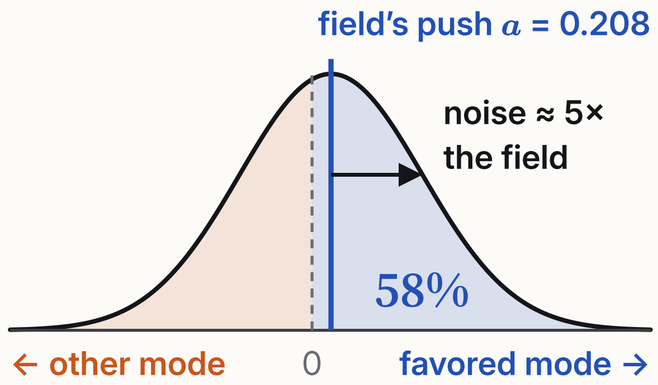


Table 4 · strongest field · after reheating

I Curie–Weiss: the exact chain tests the law

1 A weak field races noise.

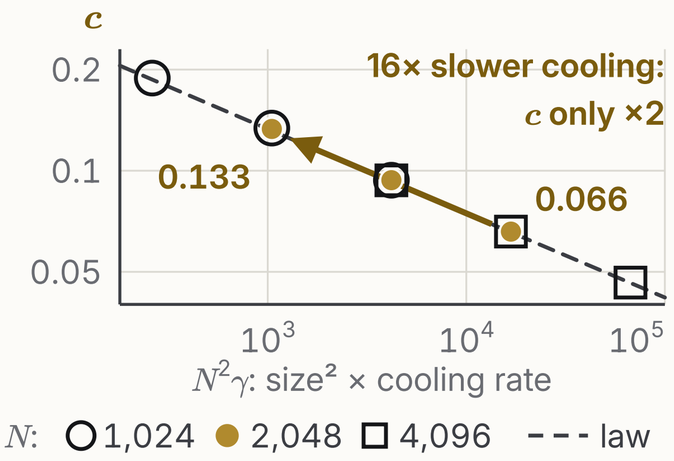


$$q_{\text{sel}} \simeq \Phi(a), \quad a = G\sqrt{N}\,(\pi/\gamma)^{1/4}$$

classical result (Kondepudi & Nelson 1983)

q: share of runs in favored mode · G: field
Fig. 1b · N = 2,048 · cooling rate γ = 0.001

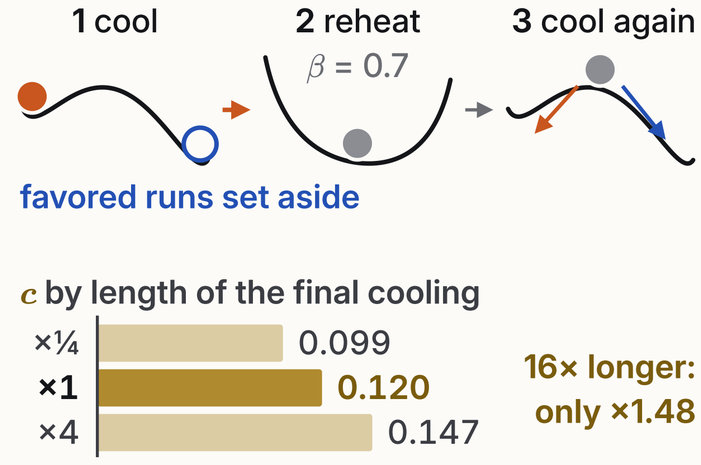
2 Slower cooling helps little.



$$c = \frac{\operatorname{logit} q_{\text{chain}}}{\operatorname{logit} q_{\text{eq}}} \propto (N^2\gamma)^{-1/4}$$

chain: sampler · eq: model · Table 2
linear ramps, NG fixed · law within 2%

3 Reheated runs reselect.

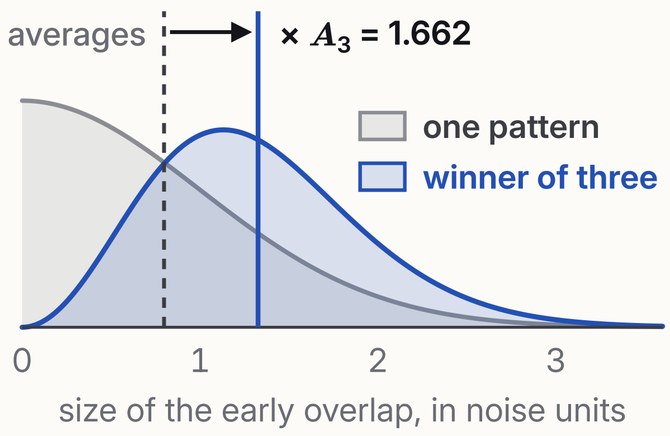


A second choice still keeps only c_{CW} = 0.120.

Table 3 · exact chain, reheated · N = 2,048

II Hopfield: competing patterns, predicted without Hopfield samples

4 Competition raises c.

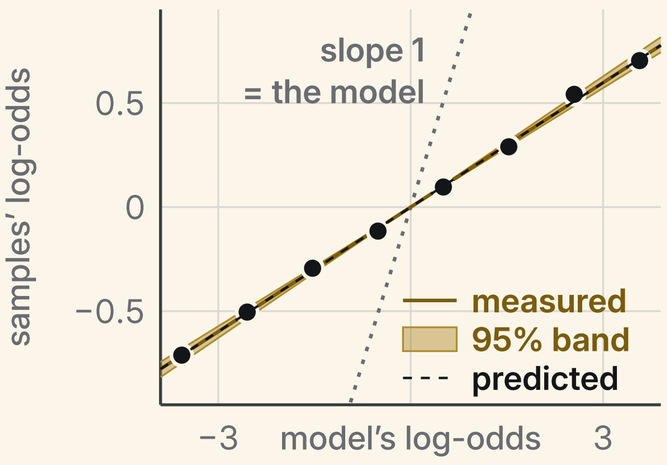


$$c_{\text{pred}} = A_p\, c_{\text{CW}}, \quad A_p = \frac{\mathbb{E}\max_\mu |Z_\mu|}{\mathbb{E}|Z|}$$

Bigger overlap, stronger bias.

Eq. 6 · Gaussian approximation

5 p = 3: 0.199, as predicted.



0.120 × 1.662 = 0.199 predicted
measured 0.199 [0.1908, 0.2095]

may partly reflect cancelling errors
Fig. 2b · 8 fields · p = 3 · N = 2,048

6 The prediction misses by up to 11%.

setting	predicted	measured	95% interval
N = 2,048, p =			
1	0.120	0.127	[0.124,0.131]
2	**0.169**	**0.188**	**[0.184,0.191]**
3	0.199	0.201	[0.194,0.210]
4	0.220	0.211	[0.191,0.239]
p = 3, N =			
1,024	0.284	0.263	[0.247,0.283]
4,096	0.150	0.160	[0.155,0.165]

Largest miss: p = 2, predicted 0.169, measured 0.188.
At p = 3, errors may partly cancel (Sec. 5).

Kondepudi & Nelson 1983 · Kondepudi & Gao 1987 · Amit et al. 1985 · Berglund & Gentz 2002 · Levin et al. 2010